

DHRUV KAURA

VOLUNTEERING

Python and Web Development Mentor

Edxcellence
Nov 2025 - Mar 2026 Remote

- Mentored 30+ students in Python backend development and web technologies; 90%+ completed hands-on capstone projects, with multiple students securing internships post-program.

EDUCATION

Bachelors of Technology in Computer Science and Engineering
GNA University CGPA: 8.15
2020-2024 Phagwara, Punjab

LANGUAGES

English	Fluent	★★★★☆
Hindi	Native	★★★★★
Punjabi	Native	★★★★★

SKILLS

Language: Python(primary), SQL, JavaScript (ES6+), TypeScript

Backend Frameworks: FastAPI, REST API Design, Asyncio, Celery, SQLAlchemy, Alembic, Django

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, HTML5 Canvas, CSS3

Generative AI & AI Tools: OpenAI APIs, LLM-based Agents, Prompt Engineering, AI Workflow Orchestration, RAG Concepts (Foundational Knowledge)

Databases: PostgreSQL, MySQL, MongoDB, Redis (caching & message broker)

DevOps & Infrastructure: Docker, Docker Compose, GitHub Actions (CI/CD), Git, Postman, Linux

Auth & Security: JWT Authentication, OAuth2, Role-Based Access Control (RBAC)

CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, Object-Oriented Programming (OOP)

Concepts: Microservices Architecture, Distributed Systems, Async Programming, System Design, API Rate Limiting, Connection Pooling, Caching Strategies

Software Engineer | GenAI Enthusiast

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SUMMARY

Backend-focused Software Engineer specializing in building high-performance APIs and scalable systems using Python and FastAPI. Experienced in designing asynchronous services, optimizing backend performance, and developing production-ready applications. Strong foundation in system design, databases, and distributed systems.

WORK EXPERIENCE

Protolabz eServices	Phagwara, Punjab
Assistant Developer	April 2024-May 2026
- Designed and deployed RESTful APIs using Python and FastAPI serving 3+ production web applications, achieving ~35% reduction in average API response time via query optimization, connection pooling, and async I/O with asyncio. - Built and maintained backend microservices integrated with WordPress and Shopify platforms, resolving 90%+ of API and CMS integration incidents within the same business day, minimizing client-facing downtime. - Containerized multi-service infrastructure using Docker and Docker Compose (API, PostgreSQL, Redis), delivering consistent parity between development and production environments with zero environment-related release failures. - Automated end-to-end CI/CD pipelines using GitHub Actions, incorporating lint checks, unit tests, and Alembic database migration workflows, cutting manual deployment effort by ~70% and eliminating regression releases.	
A2IT Online	Mohali, Punjab
ReactJS Trainee	June 2023 - Aug 2023
Independently designed, developed, and shipped a full-stack Car Service Website using ReactJS and Tailwind CSS within a 10-week program, applying component lifecycle, Context API, and client-side routing patterns.	
Cetpa Infotech Pvt. Ltd	Noida, UttarPradesh
Web Development Trainee	June 2022 - Aug 2022
Built a responsive Coffee Shop Website UI in ReactJS using Hooks-based state management, demonstrating component-driven architecture and clean separation of concerns across an 8-week accelerated program.	

PROJECTS

Real-Time API Orchestration Engine (OrchestrAI) ([Link](#))

- Architected a **distributed AI task orchestration engine** using **Python** and **LLM-based agents**, supporting **200+** concurrent API workflows with branching execution, fault tolerance, and sub-second scheduling latency.
- Improved **pipeline execution performance** by **~40%** through migration from synchronous request handling to fully **asynchronous processing** using **asyncio**, **coroutines**, and non-blocking I/O.
- Developed modular **YAML/JSON-driven workflow configurations**, reducing new pipeline onboarding time from several hours to under 15 minutes while improving scalability and maintainability.

Adaptive Workflow Intelligence System ([Link](#))

- Built a **production-grade asynchronous backend** using **FastAPI** and **PostgreSQL** with JWT authentication, RBAC middleware, and full audit logging.
- Integrated **Redis** and **Celery** for distributed background task processing with automatic retry on failure and real-time task status tracking, fault-tolerant job execution.
- Deployed complete infrastructure via **Docker Compose**; implemented **GitHub Actions CI/CD**, test execution, and Alembic migration workflows.

STRENGTHS

Analytical Problem Solving

Demonstrated through designing and implementing scalable backend systems, debugging complex application issues, and optimizing API performance for production environments.

Effective Communication

Collaborated closely with cross-functional teams, including frontend developers, product stakeholders, and clients, ensuring smooth project delivery and alignment with business requirements.

Technical Adaptability

Quickly learned and applied modern technologies such as FastAPI, React, PostgreSQL, Redis, Docker, and AI workflow tools to build and deploy high-performance applications.

Expense Splitter Backend System ([Link](#))

- Developed a multi-user **RESTful API** in **FastAPI** with group expense management, real-time balance calculation engine, and optimized SQL queries handling complex **N-party split logic**.
- Debugged and resolved **3 critical race conditions** involving transaction atomicity and concurrent balance updates, achieving **100% data consistency** across all test scenarios using database-level locking.
- Implemented **JWT-based authentication** with token refresh, RBAC, and endpoint-level authorization middleware, securing all API routes against unauthorized access.

CERTIFICATIONS

Certifications:

- Udemy Full Stack Development by Angela Yu
- AWS Academy Graduate- AWS Academy Cloud Foundation
- UpGrad Python Programming Bootcamp

Publications

- Fruits and Vegetable Sanitization and Washing Device(*Dhruv Kaura, Parul Parbhakar, Harjasleen Singh, Dr. Syed Mufassir Yaseen*)